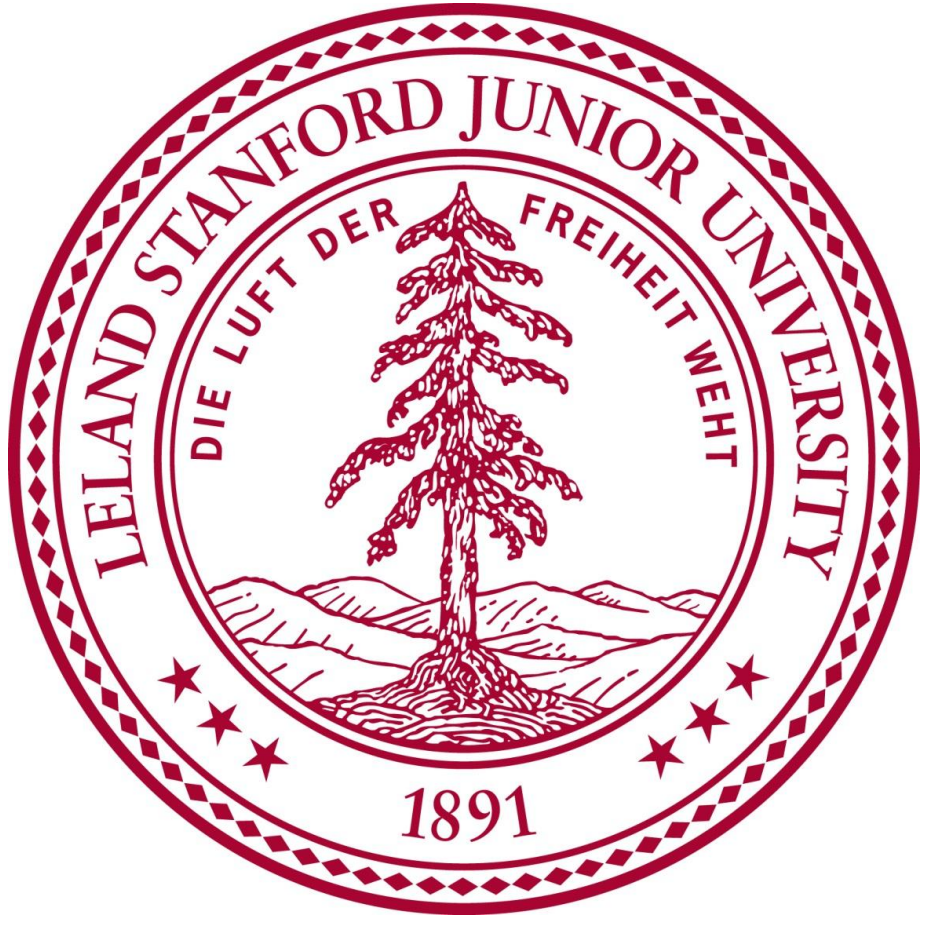




Pileup Identification for the XCC $\gamma\gamma$ Higgs Factory

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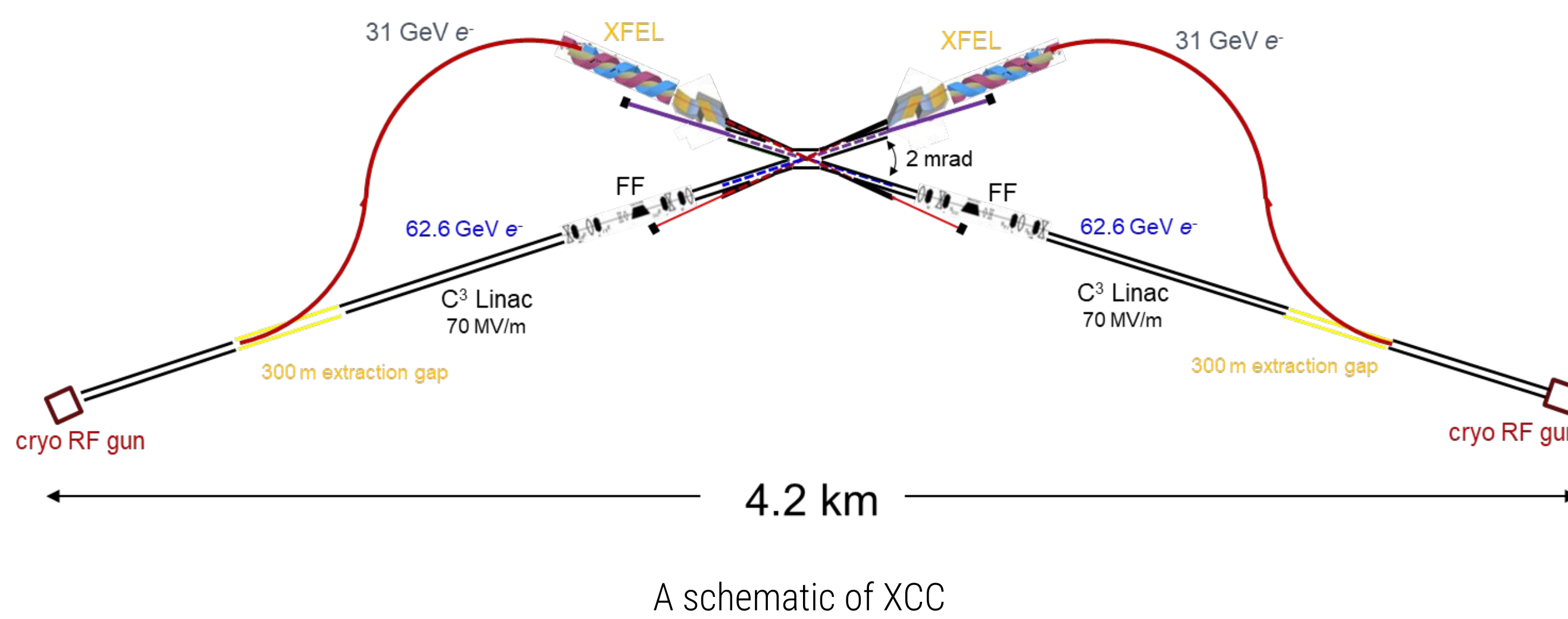
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Introduction

Historically, advances in particle physics have driven technological innovations across all fields. Take, for instance, the development of superconducting magnets and ultra-sensitive detectors, which have greatly influenced the fields of medical imaging and radiation therapy. The X-ray FEL-based $\gamma\gamma$ Compton Collider (XCC) Higgs Factory represents a transformative step forward for fundamental physics, particularly in its ability to efficiently generate Higgs bosons.

By utilizing X-ray lasers instead of traditional optical lasers, XCC is able to offer several advantages, including a lower cost, smaller footprint, and the ability to commence operations sooner compared to other proposed colliders like the International Linear Collider (ILC). Operating at a center-of-mass energy ($\sqrt{s}$) of 380 GeV makes XCC well suited to study the Higgs self-coupling via $\gamma\gamma \rightarrow HH$ processes, boasting a cross-section twice that of the ILC's $e^+e^- \rightarrow ZHH$ at $\sqrt{s} = 500$ GeV.



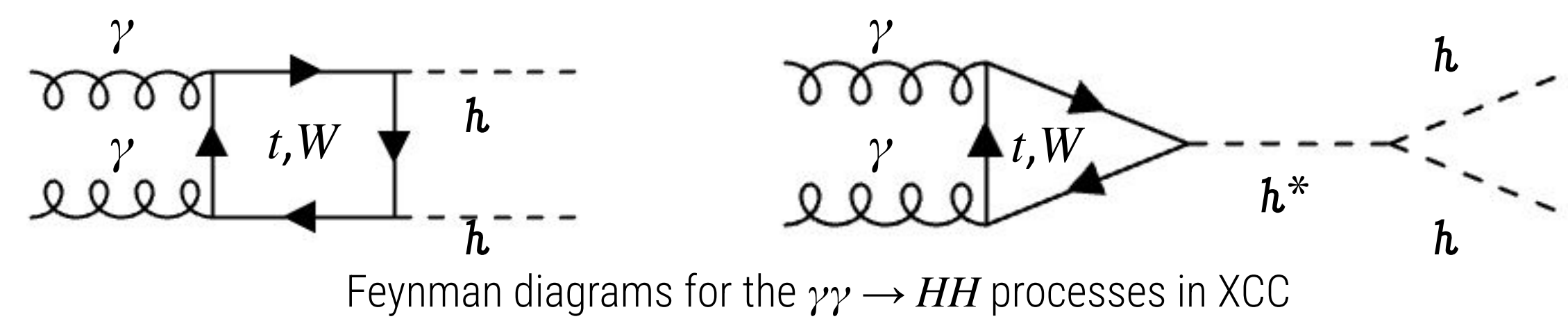
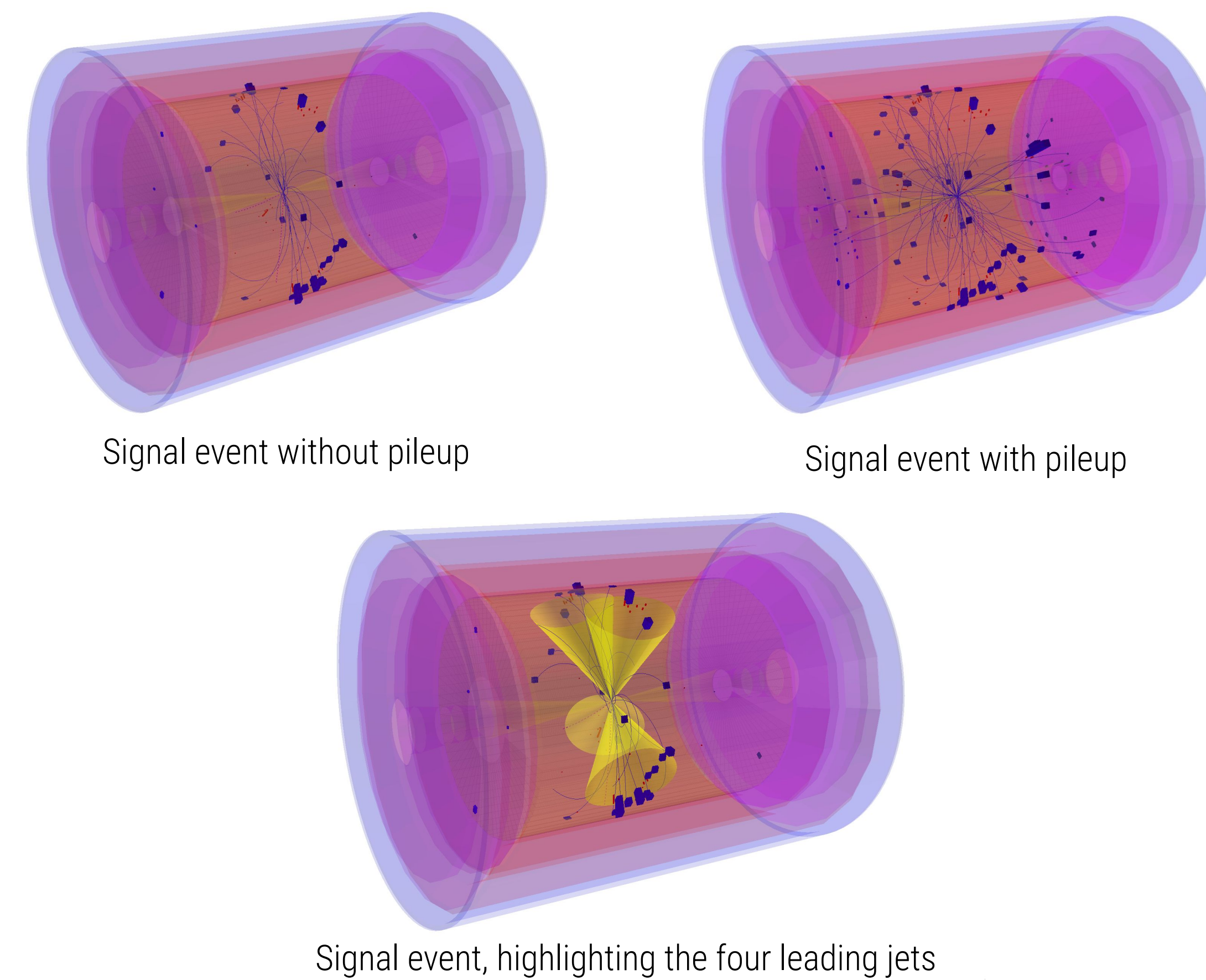
Background

XCC's design incorporates a cryogenic RF gun to produce polarized electrons, which are then accelerated by a linear accelerator using cold copper distributed coupling (C³) technology. Every other electron bunch is diverted to an X-ray Free-Electron Laser (XFEL), where a 1 keV X-ray is produced. At the Compton interaction point, these X-rays collide with the electron beam, generating gamma photons that interact to produce Higgs bosons with a mass of 125 GeV.

The unique beam conditions surrounding the XCC result in additional interactions (pileup), including beamstrahlung radiation and $\gamma\gamma \rightarrow$ hadrons. These pileup backgrounds increase the complexity of the environment and necessitate this study of their impact.

Methodology

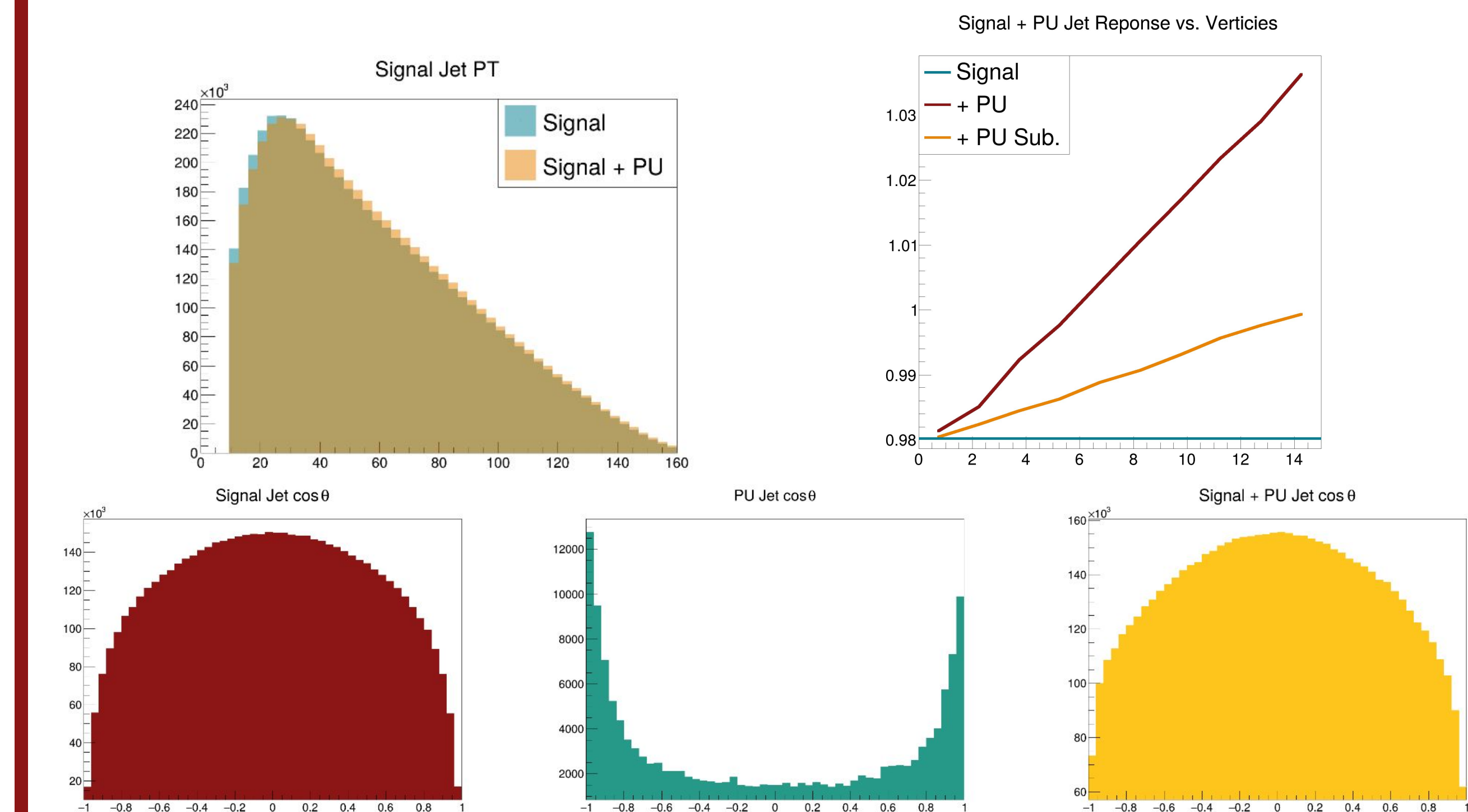
The primary goal of this research is to evaluate the impact and significance of pileup, particularly how these events contribute to uncertainty or introduce bias. To simulate and analyze the data, we utilized ROOT, a data analysis library developed by CERN, alongside Delphes, a C++ framework for fast detector simulation. We generated 1,500,000 di-Higgs signal events, followed by a second set of events with added pileup. The pileup was modeled with an average of 6.81 events per bunch crossing. Delphes simulates pileup by distributing these events along the beamline according to a Gaussian distribution, replicating the realistic distribution of collision points along the beam direction. The effect of pileup is the introduction of additional particles that contaminate the event. To account for this, we added a technique that subtracts the pileup energy event by event.



To identify the Higgs boson, we cluster particles using a jet reconstruction algorithm. Since each Higgs produces two jets, combining the four leading jets allows us to affirm the di-Higgs event. There are many different choices for jet algorithms. Similar to the Large Hadron Collider (LHC), we employ the anti- k_T clustering algorithm, particularly suitable for events with pileup backgrounds. Since anti- k_T clusters hard jets first, it effectively reduces the likelihood of pileup contamination in the jets, making it the optimal choice for this analysis.

Results

The introduction of pileup generally increases the number of reconstructed jets, although the transverse momentum (p_T) of the top four jets barely changes. As the number of vertices (pileup events) rises, there is a corresponding, albeit modest, increase in both the number of jets and their p_T . Additionally, the jet response, defined as the ratio of reconstructed jet energy to truth jet energy, gradually increases as the number of vertices grows. This is to be expected since the number of constituents in each jet should increase as more pileup events generate more particles. By applying pileup subtraction, we are able to mitigate these effects for all variables, returning the jet characteristics closer to those observed without pileup.



Analysis reveals that pileup is primarily concentrated in the forward region, dominating in the $|\cos \theta| > 0.8$ region, sometimes forming low p_T jets. Overall, the impact of pileup on the signal data is manageable. Based on our findings, we recommend the removal of all objects with $|\cos \theta| > 0.8$ to minimize pileup contamination while preserving the signal.

